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SPECIES DISCRIMINATION AND EVOLUTIONARY MODE OF *BUCHIA* (BIVALVIA: BUCHIIDAE) FROM UPPER JURASSIC–LOWER CRETACEOUS STRATA OF GRASSY ISLAND, BRITISH COLUMBIA, CANADA

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Abstract: Buchiid bivalves are geographically widespread in Upper Jurassic and Lower Cretaceous strata of the Northern Hemisphere. They are often abundant and their short stratigraphic ranges make them ideal biostratigraphic index fossils; these characteristics also render them useful for study of evolutionary patterns. We used multivariate methods to determine if we could discriminate between species of *Buchia* and examine how morphological characters change through time within the genus. Using ten morphological characters to describe shell shape and size, we tested for taxonomic differences and morphologic change in populations of buchiids collected from a single stratigraphic section on Grassy Island, located along the west coast of Vancouver Island, British Columbia. Morphometric analysis utilized traditional morphological metrics and techniques, including linear and angular measurements as well as Fourier (outline shape) analyses. Phenetic discrimination revealed considerable overlap among the recognized species in the morphospace, as well as a fairly low discriminatory power between species when compared as a group using a step-wise canonical variate analysis. Step-wise discriminant analyses between species pairs gave rise to much higher classification rates, suggesting that different characters are important for distinguishing between different species pairs. Our results also indicate that single individuals and small sample sizes of *Buchia* specimens are insufficient for biostratigraphic discrimination (unless other rarely preserved features such as the hinge and bysuss ear are available) and that a number of previously described species variants may not be taxonomically valid. A biolog using the multivariate axis that best discriminates between species (CV1) and a random walk-based test using a Hurst estimate analysis indicate a gradualistic evolutionary mode for the *Buchia* species of Grassy Island. Shell shape and size of buchiids do not appear to be closely tied to lithofacies changes over the *c.* 10 myr time interval, suggesting that ecophenotypic variation (as it relates to substrate changes) probably had minimal influence on morphology.

Key words: biostratigraphic zones, gradualism, multivariate morphometrics, taxonomy.

THE genus *Buchia* Rouillier, 1845 is among the most biostratigraphically important cosmopolitan genera of the later Mesozoic (Kauffman 1973) and has been used traditionally to correlate Upper Jurassic–Lower Cretaceous strata of the Northern Hemisphere. Species of *Buchia* are biostratigraphically valuable because they are abundant in a variety of facies, widely distributed geographically, characterized by a high rate of evolutionary turnover, and readily determined taxonomically by focusing on morphological characteristics such as shell sculpture and shape, beak incurvature, and shape of the bysuss ear (Jeletzky 1965). Assessing evolutionary trends within the genus, Jeletzky (1965) proposed a succession of biostratigraphic zones utilizing species of *Buchia* for the western Canadian Cordillera, British Columbia. These established *Buchia* zones are widely used today in geological correlation and include the *B. columbiana* Grey *et al.*, 2008, *B. okensis* Pavlow, 1907, *B. uncitoides* Pavlow, 1907, *B. tolmatschowi* Sokolov, 1908 and *B. pacifica* Jeletzky, 1965 zones (Jeletzky 1965; Grey *et al.* 2008).

Morphological features such as the bysuss ear and parts of the right ligamental plate change considerably in buchiids over geologic time, yet these features are rarely preserved and are difficult to describe morphometrically (Jeletzky 1965, 1966); therefore, it is critical to be able to identify specimens even when these features are not available (which is often the case). More easily recognizable are changes in the beak orientation and overall shape of

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the shell (i.e. retrocrescent, infracrescent and straight forms), and these are the features we focus on for this paper.

Despite being biochronologically valuable, disagreements in the designation of buchiid species within the genus are common. Imlay (1959) was the first to suggest that the genus had been subdivided into a biostratigraphically unrealistic number of species on the basis of small differences in shell shape, size and ornamentation. Imlay (1959) also noted that many of the previously recognized species grade into each other within a single population and he attempted to take into account biological and environmental factors affecting the variability of the shell morphology. In contrast, Jeletzky (1965) disagreed with Imlay's amalgamation of species and followed the nomenclature originally proposed by Crickmay (1925, 1930) and others (Pavlow 1907; Sokolov 1908) who not only split buchiids into many more species, but also named several variants. These variants include *B. okensis* var. *subokensis* Pavlow, 1907, var. *canadiana* Crickmay, 1930, and var. *elliptica* Pavlow, 1907; *B. uncitoides* var. *catamorpha* Crickmay, 1930, var. *spasskensoides* Crickmay, 1930, and var. *acutistriata* Crickmay, 1930; and *B. tolmatschowi* var. *americana* Sokolov, 1908. These examples illustrate the need for objective, statistical testing for species discrimination in this fossil taxon.

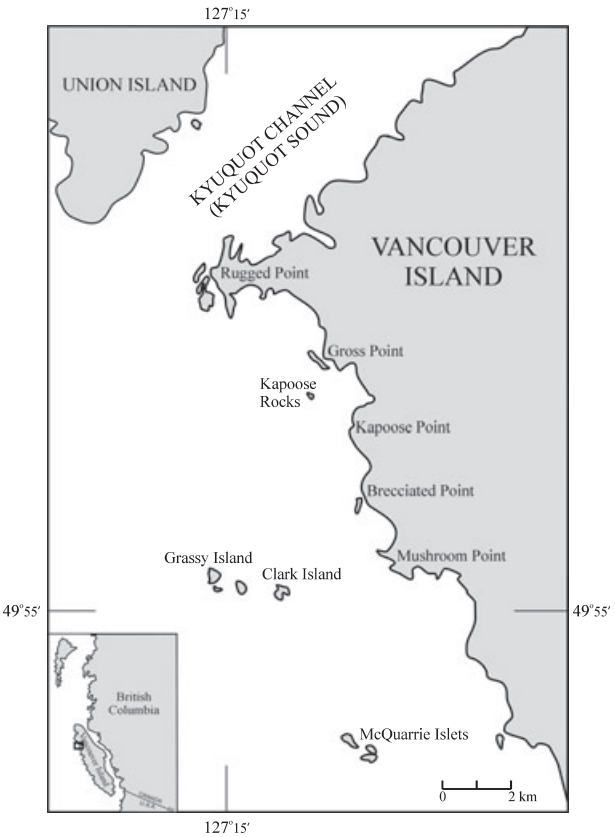
Today, over 30 species are assigned to the genus *Buchia* (Zakharov 1981) and the reliability of species designations has never been verified using the now common practice of morphometrics. Despite their importance for biostratigraphy, there have been few, if any, quantitative studies of buchiid bivalves (but see Crampton 1994) and even standard measurement parameters for the valves have not been established.

The purpose of this study is two-fold. First, we develop a protocol for the systematic description of *Buchia* shell morphology using multivariate morphometrics and employ this to examine species designations. The biostratigraphic value of the *Buchia* succession is contingent on the ease of taxonomic differentiation of the various species in the lineage and this should be rigorously verified through quantitative study. Second, we assess the evolutionary mode of *Buchia*, based on a large number of populations from a single stratigraphic section, with a consideration of whether ecophenotypic variation is an important influence on morphology.

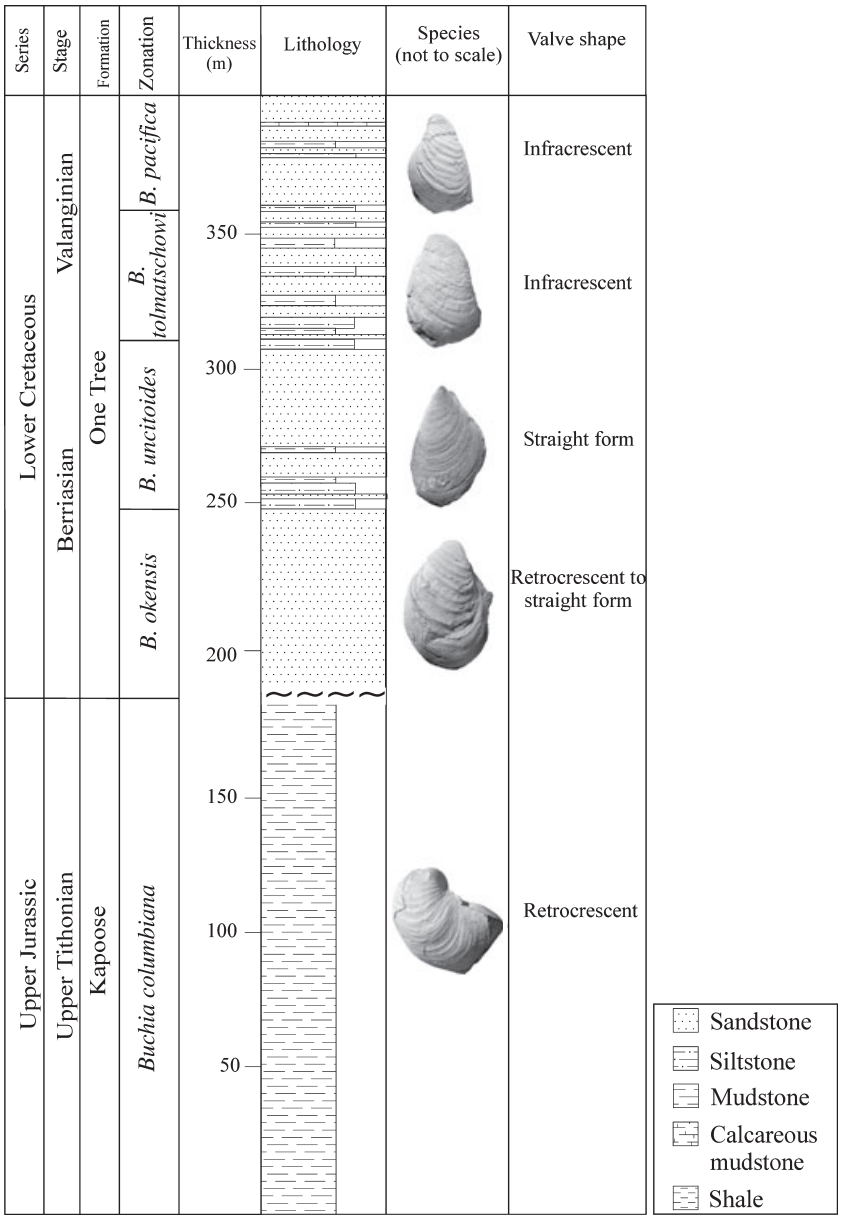
MATERIAL AND METHODS

Material

Buchiid material used in this study, mostly consisting of casts and internal moulds of shells, comes from collections of the Geological Survey of Canada (GSC) held in Ottawa, Ontario, and Vancouver, British Columbia. The collections were originally made by J. A. Jeletzky in the 1940s–60s during several visits to Grassy Island off the west coast of Vancouver Island (Text-fig. 1). The Grassy Island section spans the Tithonian (Upper Jurassic)–Lower Valanginian (Lower Cretaceous) and includes the upper part of the Upper Jurassic Kapoose Formation of Muller *et al.* (1981) and the lower part of the Lower Cretaceous One Tree Formation of Bancroft (1937) and Jeletzky (1950, 1965) (Text-fig. 2). The Kapoose Formation (Tithonian) consists of mudstone and shale beds with abundant calcareous concretions and common sandstone interbeds, whereas the One Tree Formation (Berriasian–Valanginian) consists primarily of medium- to coarse-grained sandstone beds with interbeds of siltstone and sandy shale (Grey *et al.* 2007). On the basis of missing faunal zones, a disconformity spanning the uppermost Tithonian is inferred to separate the two formations (Jeletzky 1965). These missing zones include *B. piochii* f. typ. Gabb, 1864, *B. fischeriana* d'Orbigny, 1845 and *B. terebratuloides sensu lato* Lahusen, 1888. Both formations are assigned to the Middle Jurassic–Lower Cretaceous Kyu-



TEXT-FIG. 1. Location map of the stratigraphic section on Grassy Island, off the west coast of Vancouver Island, British Columbia (NTS map-area 92E/14).



TEXT-FIG. 2. Stratigraphic column of the Jurassic–Cretaceous succession exposed at Grassy Island, with key *Buchia* species photographs and valve shape information found within each zone. Bed-by-bed fossil collections were made throughout the entire section at Grassy Island and were initially identified by J. A. Jeletzky. Details of the stratigraphic distribution of fossil collections are presented in Grey *et al.* (2007).

quot Group (Muller *et al.* 1981) and are interpreted as a shallow-marine shelf assemblage characterized by extensive storm deposits.

Jeletzky collected bed-by-bed through the Grassy Island section, and the resultant collections provide a large number of stratigraphically well-constrained populations of buchiid specimens and other associated molluscs. The collections are from 113 different stratigraphic levels, each representing a single bed. Beds range in thickness from 0.06 to 24 m, with a mean thickness of 1.95 m. Jeletzky made a fossil collection from each bed and each collection includes dozens to hundreds of individual buchiid specimens. Each collection thus provides a representative overview of buchiid species present during a single 'short' time interval.

Fifteen different species and variants of *Buchia* have been documented by Jeletzky in the Grassy Island section, but the principal taxa are *B. columbiana*, *B. okensis*, *B. uncitoides*, *B. tolmatschowi* and *B. pacifica*. Shape trends in the left valves can be easily recognized within the section: *B. columbiana* are retrocrescentic (sickle-shaped; Newell and Boyd 1970); *B. okensis* are retrocrescentic to straight; and *B. uncitoides*, *B. tolmatschowi* and *B. pacifica* are straight to infracrescentic (Text-fig. 2). For a detailed summary of the Grassy Island section, see Grey *et al.* (2007).

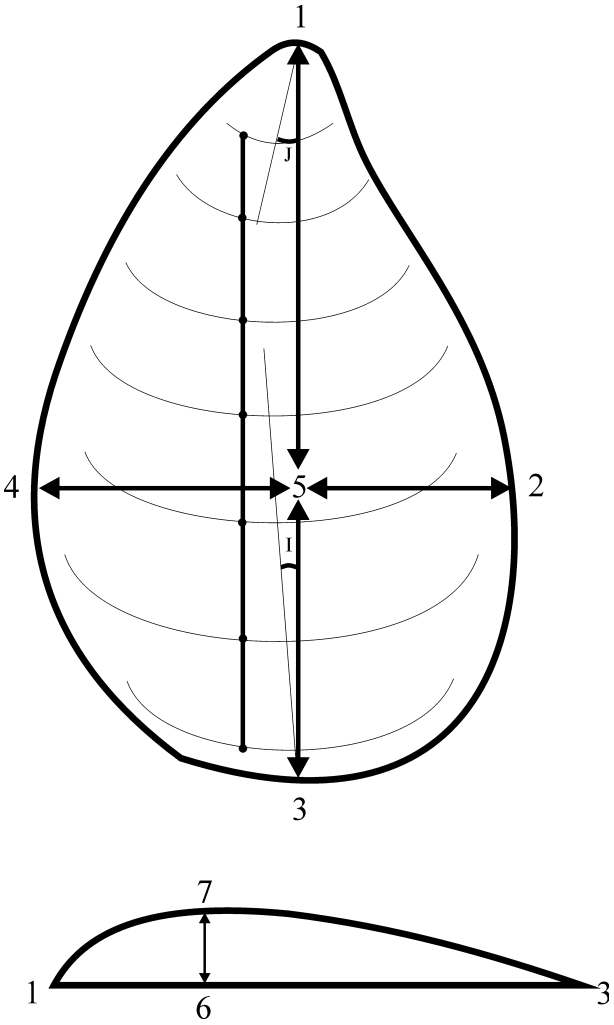
Using the Jeletzky collections, we measured a total of 765 shells selected from 113 successive stratigraphic levels within the section. The shells range in length from 10 to 70 mm. In addition, we assigned collections to a lithofa-

cies (calcareous mudstone, mudstone, sandstone, shale and siltstone) based on an examination of the matrix, as well as Jeletzky's descriptive comments and our personal field observations.

Morphometric measurements

The exterior and side views for all unbroken left valves from each stratigraphic level were photographed in the same orientation using a Nikon D70 digital camera. We used the left valve for consistency and also because it exhibits greater morphological variance, similar to inoceramids (Crampton 1996). Images were saved to computer files and measured with a custom-designed morphometrics program (MorphLab 1.0).

To describe shell shape, we used ten morphometric variables that include angular and linear measurements and one calculation of the average distance between commarginal (or quasi-marginal; Ubukata 2005) ribs (Text-fig. 3; for variable means for each species, see Appendix). These ribs are not necessarily related to growth lines and several mechanisms have been proposed for their formation, including oscillation in regulatory systems (Hammer 2000). Of particular note is the variable termed the 'crest-line': this is defined as the imaginary line (corresponding to the maximum growth vector) joining the highest points of the left valve of *Buchia* (Pavlow 1907) and has been recognized as a taxonomically valuable feature (Jeletzky 1966). We assessed the taxonomic significance of these morphometric variables, allowing us to establish a statistical protocol for the discrimination of *Buchia* species. Such variables should be mutually independent (Schatz 2001) and should contribute to the overall discrimination of species. We explored the extent to which this is the case in the taxonomic analyses presented below. Outline shapes, defined as a series of points around the perimeter of the shell, were also generated by the MorphLab 1.0 program and were used for Fourier analyses.



TEXT-FIG. 3. Features used to characterize shell morphology in buchiid bivalves: 1, dorsalmost point (umbo) on hinge margin; 2, posteriormost point; 3, ventralmost point; 4, anteriormost point; 5, intersection of lines measuring maximum valve length and valve width; 6, point directly beneath highest point on left valve; 7, highest point on left valve. Morphological variables measured: ventral angle of crest-line (I); dorsal angle of crest-line (J); dorsal length (Ld; 1–5); ventral length (Lv; 5–3); anterior width (Wa; 4–5); posterior width (Wp; 5–2); dorsal distance to highest point on valve (Dd; 1–6); ventral distance to highest point on valve (Dv; 6–3); inflation (In; 6–7) and delta (Δ; average distance between commarginal rings, represented on the figure as dots on the rings with a line through them).

Taxonomic analyses

We utilized phenetic discrimination of linear and angular measurements (Kowalewski *et al.* 1997) and outline shape (Fourier analyses: Crampton 1996; Haines and Crampton 2000) to define the *Buchia* morphospace and determine morphospecies. Linear and angular variables of buchiids were log-transformed prior to analysis (Kowalewski *et al.* 1997). Landmark-based statistical methods, although more powerful, could not be employed in our study because of a lack of homologous points on the *Buchia* shell. Despite this limitation, phenetic discrimination using traditional morphometric techniques has proven useful in providing taxonomic resolution even for morphologically simple fossils (e.g. lingulide brachiopods; Kowalewski *et al.* 1997) and sister species (e.g. arcid bivalves; Marko and Jackson 2001).

Phenetic discrimination included a combination of principal component analyses (PCA) and a step-wise canonical variate analysis (CVA). PCA is a technique that condenses information from many correlated variables into a few quantities that summarize the variation among

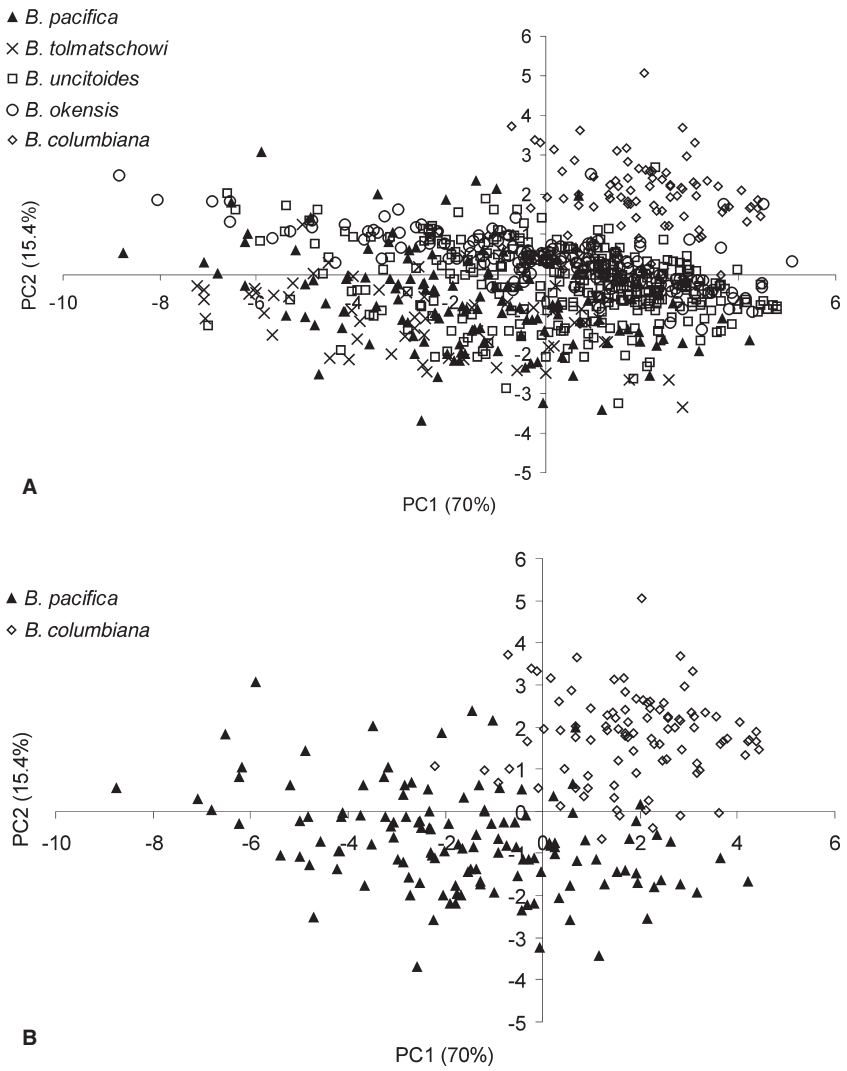
TABLE 1. Correlation of all morphological variables on the first two principal components: variables are defined in Text-fig. 3.

Variable	PC1	PC2
Dorsal angle of crest-line (J)	−0.33	0.86
Ventral angle of crest-line (I)	0.42	−0.82
Dorsal length (Ld; 1-5)	0.95	0.13
Ventral length (Lv; 5-3)	0.95	0.04
Anterior width (Wa; 4-5)	0.92	0.22
Posterior width (Wp; 5-2)	0.90	0.08
Dorsal distance (Dd; 1-6)	0.93	−0.05
Ventral distance (Dv; 6-3)	0.91	0.23
Inflation (In; 6-7)	0.93	0.05
Delta	0.84	−0.02

individuals; it was used to explore the *Buchia* morphospace and was executed using the program PAST (version 1.38; Hammer *et al.* 2001).

The CVA was used to determine which variables discriminate between groups of previously identified species and also for predictive classification. *A priori* grouping for the canonical variate analysis was based on Jeletzky's taxonomic identifications (Grey *et al.* 2007). We utilized a step-wise method for the CVA because it selects variables that contribute the most discriminatory power to the model (refer to Cheetham *et al.* 2006); this method is advantageous when a number of characters are measured. Resampling methods help to assess the accuracy of analyses by testing the repeatability of the results; we therefore simulated a jack-knifed approach in SPSS (version 11.0) by selecting the 'leave-one-out classification' option. For comparison, we ran ten step-wise discriminant analyses (DA) on all species pairs and applied a Bonferroni correction for multiple tests. Discriminant analyses are similar to CVAs but are used when comparing only two groups.

Species were recognized statistically if the PCA showed distinctive morphospaces and/or the CVA/DAs indicated



TEXT-FIG. 4. A, PCA results show overlap in *Buchia* morphospace for all species based on visual assessments by Jeletzky from Grassy Island. Axes PC1 and PC2 represent 70 per cent and 15.4 per cent, respectively, of the character variation within the section. B, PCA results, showing clear separation in the morphospace for *B. pacifica* and *B. columbiana*.

a reasonably high discriminatory power (i.e. a high percentage of individuals that correctly match Jeletzky's taxonomic identifications). A low percentage of individuals that match Jeletzky's classification indicate that either species assessments based on visual examination may be inaccurate or our statistical tests may be missing important taxonomic information. In order to explore the effect of ontogenetic variation, we also performed analyses by size class based on valve length: 10–30 mm (juvenile); 30–50 mm (immature); and 50–70 mm (mature).

Outline shape can be a valuable morphological character, making Fourier analyses ideal for morphometric studies, especially for bivalves lacking homologous points on their shells (Crampton 1995; Crampton and Maxwell 2000). Analyses were run in Hangle (Crampton and Haines 1996) and Htree (Crampton and Haines, unpublished). Fourier coefficients were standardized for size based on shell area; therefore, the analysis is based solely on shape changes. A PCA and a CVA were performed on the resulting suite of Fourier coefficients in the same manner as outlined above, with the exception that the PCA was based on the variance-covariance matrix (Crampton 1995).

Evolutionary mode

We explored evolutionary mode using two methods. A biolog, produced by plotting the first canonical variate (CV1) values against stratigraphic position, was created as it helps to illustrate temporal morphologic patterns (Kelley 1983; Reyment 1980). We employed Roopnarine's (2001) random walk model for a stratophenetic series as a quantitative method for exploring evolutionary mode. Tests of evolutionary mode in a stratophenetic series require use of the random walk model as the null hypothesis because random walks can display characteristics indicative of directional evolutionary trends (Roopnarine 2001). Roopnarine's (2001) method calculates the Hurst estimate (h) of a series and is based on the Hurst exponent (H), a measure of its persistent, or anti-persistent, deviation from any overall trend; see Roopnarine *et al.* (1999) and Roopnarine (2001) for further details. The Hurst exponent is estimated by regressing all log-transformed pair-wise differences of morphological values against the relevant log-transformed intervals of relative or absolute time (Roopnarine 2001). Morphological values in this case are the averaged primary canonical variate (CV1) values for each stratigraphic level calculated from the morphometric analysis as described earlier. We utilized the random walk program, Enigma-2.4 for Linux (Roopnarine 2001), to assess evolutionary mode within the Grassy Island section. Hurst estimate values can range from 0 to 1; high values (i.e. $h > 0.9$) indicate strong directional change, whereas low values can indicate stasis ($h < 0.2$) or random walks ($0.25 < h > 0.55$) (Roopnarine 2001). P values indicate the probability of a random walk: low P values ($P < 0.5$) mean that there is a small probability the pattern was generated by a random process, while high P values ($P \geq 0.9$) indicate random walks (Roopnarine 2001).

Correlation between lithology and morphology

We performed a principal component analysis with lithology as the grouping variable to assess the correlation between changes in shell morphology and lithology (a proxy for palaeoenvironment). If the data group by lithology, then morphological variation is dependent on environmental conditions, suggesting that ecophenotypic variation is an important factor controlling morphology. This is a coarse study of ecophenotypy as we are only assessing how it relates to substrate changes and do not take into account biological factors (e.g. predation pressure, population density), nor do we include a comprehensive facies analysis.

RESULTS AND DISCUSSION

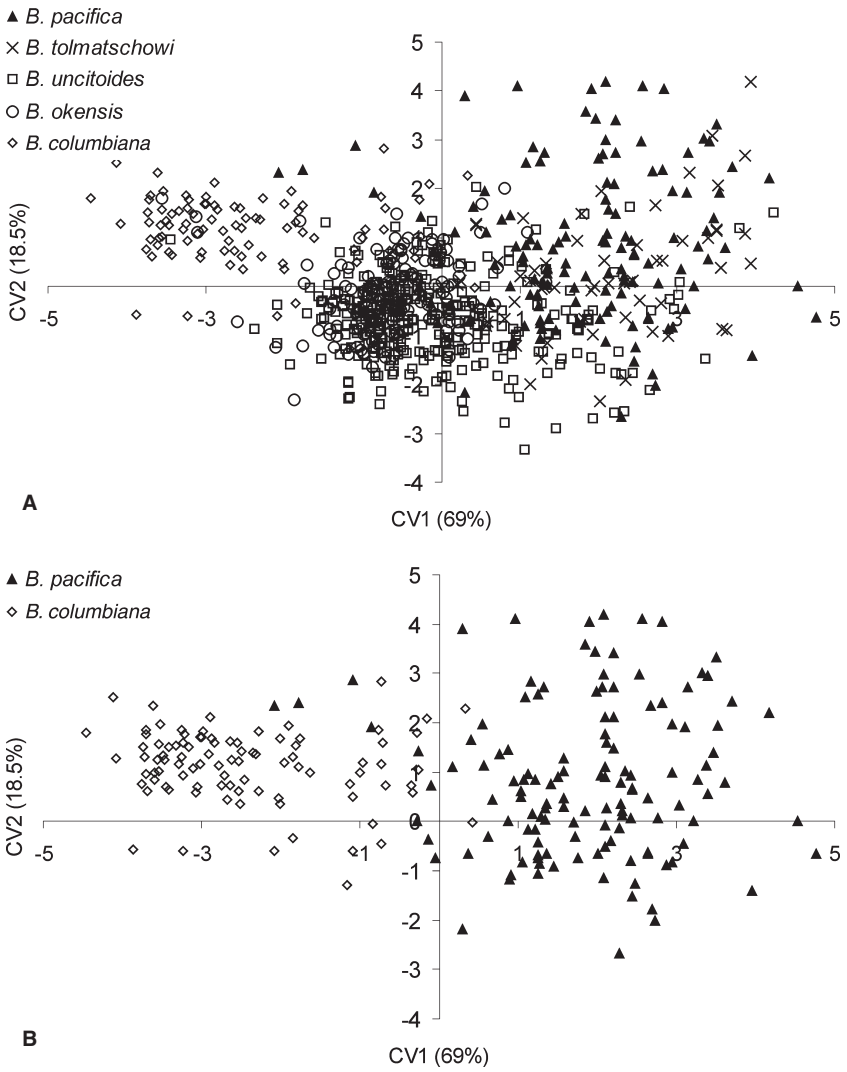
Taxonomic analyses

The primary principal component axis (PC1 = 70 per cent of variation) was a function of the linear measure-

TABLE 2. Classification summary for specimens from the Grassy Island section, contrasting the original with the cross-validated (based on a simulated jack-knifed approach) results.

Species	Predicted group membership					Total
	1	2	3	4	5	
Original						
1	81 (85)	11 (12)	2 (2.1)	0 (0.0)	1 (1.1)	95
2	2 (1.3)	103 (69)	42 (28)	1 (0.70)	2 (1.3)	150
3	6 (1.8)	67 (20)	222 (68)	17 (5.2)	17 (5.2)	329
4	0 (0.0)	1 (1.7)	6 (10)	47 (78)	6 (10)	60
5	3 (2.3)	5 (3.8)	17 (13)	40 (31)	66 (50)	131
Cross-validated						
1	80 (84)	11 (12)	3 (3.2)	0 (0.0)	1 (1.1)	95
2	2 (1.3)	100 (67)	44 (29)	1 (0.70)	2 (2.0)	150
3	6 (1.8)	68 (21)	218 (66)	19 (5.8)	18 (5.5)	329
4	0 (0.0)	1 (1.7)	6 (10)	44 (73)	9 (15)	60
5	3 (2.3)	5 (3.8)	17 (13)	42 (32)	64 (49)	131

Numbers in parentheses represent the percentage of correctly classified individuals: 68 per cent of the original grouped cases were correctly classified according to Jeletzky's visual assessments; 66 per cent of the cross-validated grouped cases were correctly classified. 1, *B. columbiana*; 2, *B. okensis*; 3, *B. uncitoides*; 4, *B. tolmatschowi*; 5, *B. pacifica*.



TEXT-FIG. 5. A, results from the CVA for all of Jeletzky's assigned species from Grassy Island. Axes CV1 and CV2 represent 69 per cent and 18.5 per cent, respectively, of the character variation within the section. CV1 represents crest-line variation (angles J and I on Text-fig. 3) and CV2 represents shell inflation ('In' on Text-fig. 3). B, CVA results, showing clear separation of *B. pacifica* and *B. columbiana*.

ments and delta, whereas PC2 (15.4 per cent of variation) was represented by the crest-line angles (Table 1). The principal component analysis shows considerable overlap in the morphospace between previously designated species (Text-fig. 4A), with the exception of the two end members (*B. pacifica* and *B. columbiana*) (Text-fig. 4B). A large degree of morphological variation from the inferred mode of life (crowding) of these taxa was expected and may mean that a reasonably large sample (e.g. 10–30 specimens) is required for confident assignment to species. A similar conclusion was reached by Kelly (1990) for buchiids in Europe.

Approximately two-thirds of the specimens were classified correctly according to Jeletzky's idenfications, and all four canonical variate functions are significant, but only the first two have been plotted as they account for most of the variation (87.5 per cent; Text-fig. 5A–B; Table 2). All variables were used in the step-wise canonical variate analysis, but the primary canonical variate (CV1) is largely a function of the crest-line angles. CV2 is primarily a function of inflation. The very large within-sample variances contribute to reduced discrimination.

Results that were divided by size class (not figured) showed similar separation between *Buchia* species as the results with all size classes, indicating that differences are not based on ontogenetic stage (proxied as length; see Crampton and Maxwell 2000).

Discriminant analyses between all species pairs revealed a much higher percentage of correctly classified specimens, ranging from 77–99 per cent (Table 3), indicating morphological differences between all previously identified species. Therefore, Jeletzky's visual assessment of species is statistically supported. This result also demonstrates that, depending on the species pair, different morphological variables are important for discrimination (Table 3). Therefore, while results from the CVA indicate that the angles of the crest line and inflation have the most discriminatory power, step-wise DAs between

TABLE 3. Classification summary for ten discriminant analyses using species as *a priori* groupings for specimens from the Grassy Island section.

Species pair	Per cent correct	Wilks' λ	P-value	Sample size	Characters
B. pacifica *B. tolmatschowi*	78 (77)	0.637	0.000*	191	Dd, J, In, Δ
B. pacifica *B. uncitoides*	91 (90)	0.435	0.000*	465	In, I, Dd, Lv, Wa, Ld, Dv, J
B. pacifica *B. okensis*	94 (94)	0.329	0.000*	281	I, In, Ld, Wa, Dv
B. pacifica *B. columbiana*	97.3 (97)	0.208	0.000*	226	I, J, Wp
B. tolmatschowi *B. uncitoides*	93 (92)	0.450	0.000*	389	In, Lv, J, Wp, Dv
B. tolmatschowi *B. okensis*	99 (98)	0.228	0.000*	210	I, J, Dd, Lv, Wa, Dv
B. tolmatschowi *B. columbiana*	99 (99)	0.137	0.000*	155	I, J, Lv
B. uncitoides *B. okensis*	75 (74)	0.745	0.000*	479	I, J, Wa, Wp, Δ, In, Ld
B. uncitoides *B. columbiana*	95 (95)	0.404	0.000*	424	J, I
B. okensis *B. columbiana*	96 (94)	0.279	0.000*	245	I, J, Δ, Wa, In

Numbers in parentheses represent the percentage correctly classified after cross-validation (based on a simulated jack-knifed approach across specimens). The last column indicates which characters, in order of most to least importance, were picked by the step-wise DA for discrimination between the two species. For explanation of character abbreviations, see Text-figure 3; *all tests are significant after a Bonferroni correction for ten tests ($\alpha = 0.05/10$).

species pairs demonstrate that the variables with the most discriminatory power can differ. We therefore recommend that all ten morphological variables be measured and used in subsequent taxonomic analyses of the genus *Buchia*.

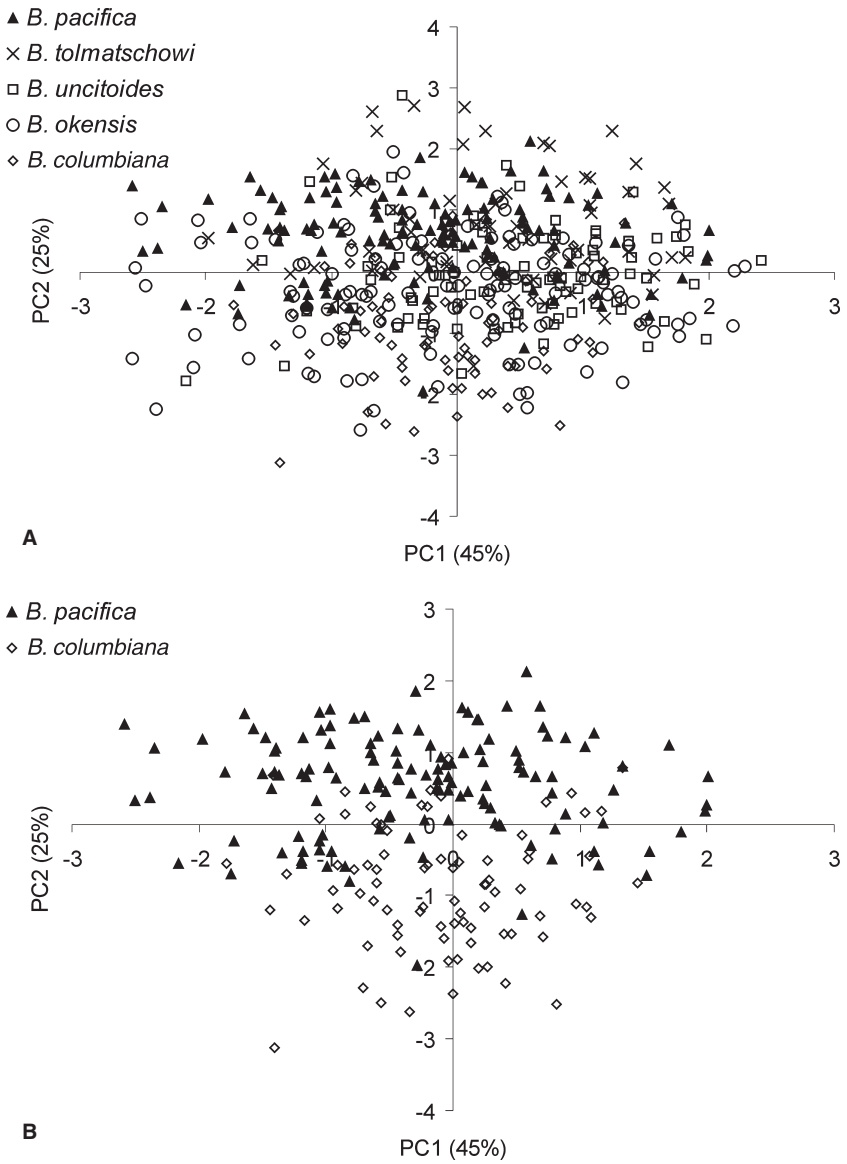
These results, taken together, show that the morphology of *Buchia* species is highly variable and that different species of buchiids will be difficult to recognize in small sample sizes (e.g. <10 specimens). This observation is corroborated by taxonomic identification of buchiid collections using non-statistical techniques. Crampton (1996) found similar results for an Albian inoceramid (*Actinoceramus*) and suggested that, despite morphological overlap between species, they could be classified as taxonomically distinct with an examination of sufficiently large sample sizes. This issue has ramifications for the usefulness of the *Buchia* zones constructed by Jeletzky (1965) and may be a more pervasive problem throughout the genus. Similar morphological analyses of other *Buchia* zones in other geographic regions are required to assess fully the implications for taxonomic differentiation within the genus.

This research is not intended to replace traditional methods of identifying buchiids, only to enhance it. When available, hinge information, right valve morphology and more detailed observations of shell ornamentation are important factors involved in identification. We show that even when these features are not preserved it is still possible to discriminate between species with large enough sample sizes. Our results also suggest that some, perhaps most, species variants may not hold a great deal of taxonomic value. We encourage the use of multivariate morphometrics, in association with details of the hinge and other characters, in such analyses because visual assessments are more subject to interpretation than are those obtained through quantitative methods.

Fourier analyses yielded results that were also suitable for discrimination of *Buchia* species (Text-fig. 6A). PC1 and PC2 captured a total of 70 per cent of the variation and 63 per cent of the specimens were classified correctly with the CVA. The results are similar to the phenetic discrimination with the linear and angular measurements in that there is no clear separation in the morphospace (Text-fig. 6A) except between *B. pacifica* and *B. columbiana* (Text-fig. 6B).

Evolutionary mode

The biolog (Text-fig. 7) created for all individuals of *Buchia* on the discriminant axis that best separated the pre-

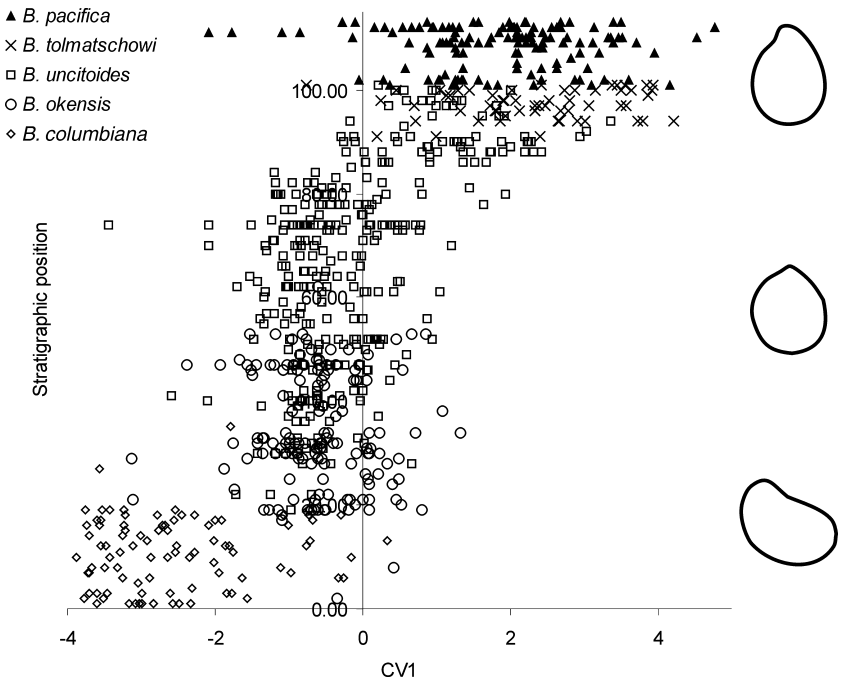


TEXT-FIG. 6. A, PCA results from the Fourier analysis, illustrating overlap in the morphospace for all of Jeletzky's assigned species from Grassy Island. Axes PC1 and PC2 represent 45 per cent and 25 per cent, respectively, of the character variation within the section. Fourier analyses captured much less of the morphological variation in buchiid shells than analyses using angles and linear measurements. B, PCA results from the Fourier analysis, showing separation in the morphospace for *B. pacifica* and *B. columbiana*.

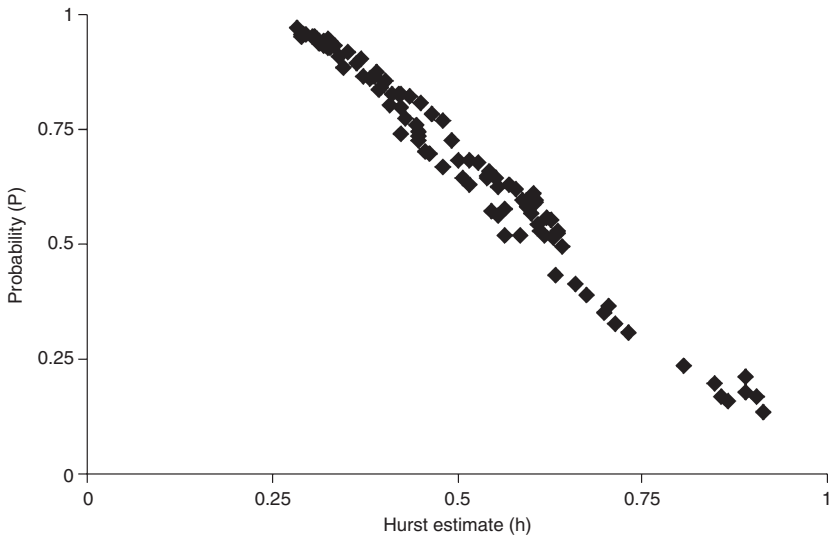
viously assigned species (CV1) shows a gradual shift from *B. columbiana* to *B. pacifica*, where differences are based predominantly on angles of the crest-line. We qualitatively classify this as a gradualistic trend because the change is consistent and directional through time and the two end members separate well from each other.

The result from the Hurst estimate analysis, using the primary canonical variate as a measure of overall morphological change, exhibits weak directionality throughout the entire interval of time. This is documented by the trajectory's high Hurst estimate values (h) and relatively low probabilities (P) of being generated by a random process (Text-fig. 8). The null hypothesis that the overall series represents a random walk is therefore rejected, supporting the qualitative conclusion that the trend is gradualistic. However, the mid-section of the biolog represents a random walk and this corresponds to low h values ($h \leq 0.5$) and high P values ($P \geq 0.9$) as calculated from the Hurst estimate analysis. We explain the apparent large jump in morphology between the *B. columbiana* and *B. okensis* zones as related to the disconformity in the section (Text-fig. 2; Jeletzky 1965) rather than a punctuated change.

As far as we are aware, this is the first study to apply Roopnarine's (2001) random walk test to macrofossils. Future research will ascertain whether the gradualistic evolutionary pattern is common within the buchiid lineage by comparing modes from a variety of geographic settings. Some members of the closely related family Inoceramidae also display gradual evolutionary modes (e.g. Crampton 1996; Crampton and Gale 2005) and other Mesozoic bivalves display non-branching gradual-



TEXT-FIG. 7. Biolog created by plotting the canonical variate means (centroids) for the first canonical variate (CV1) against stratigraphic position. Outlines on the right are plots of the average shape of the shells as they relate to the morphospace.

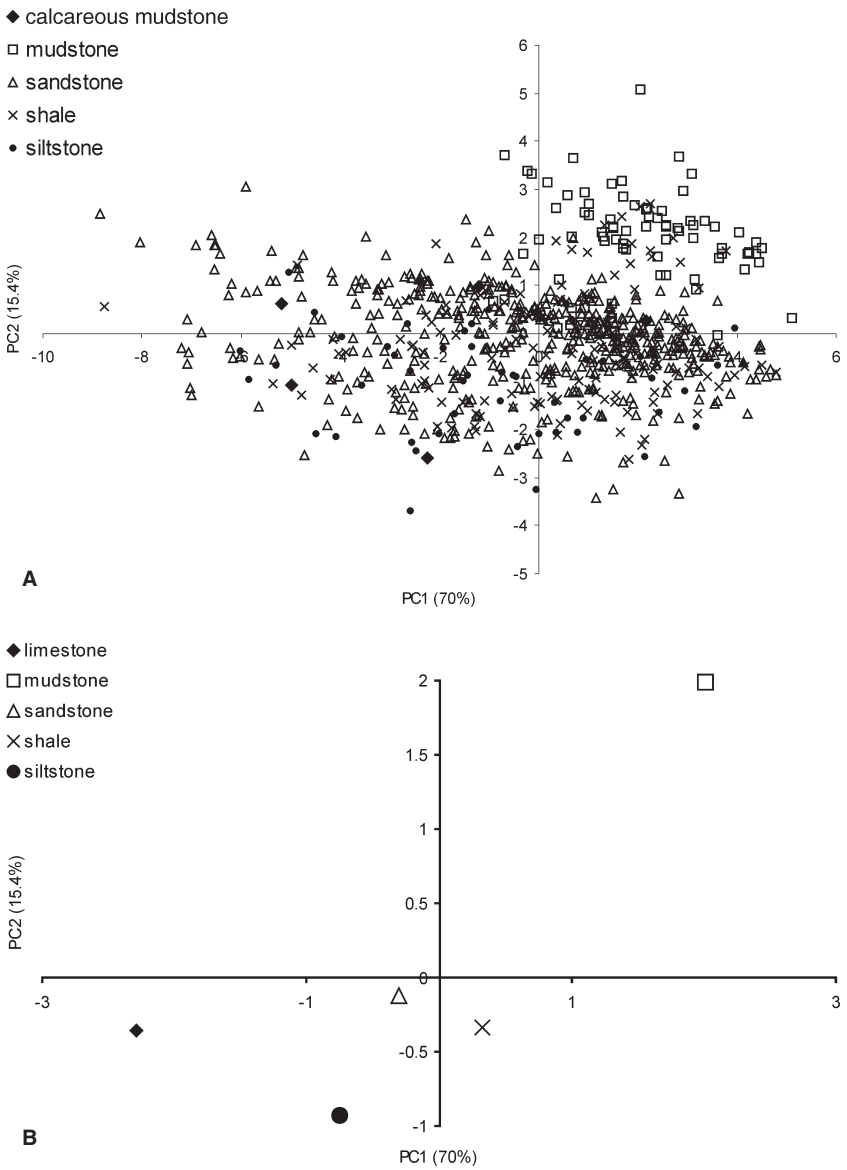


TEXT-FIG. 8. Hurst estimate analysis of the evolutionary pattern described by the stratophenetic series, with h values >>0.5 indicating increasing directionality and a gradualistic evolutionary mode.

ism (Fortey 1988) and gradualism and stasis (Geary 1987).

Correlation between lithology and morphology

The PCA with linear/angular measurements using lithology as the grouping variable shows no clear groupings (Text-fig. 9A), nor does there appear to be a trend in mean shape from coarse to fine facies (Text-fig. 9B). The notable exception is the separation between mudstone and all other lithofacies, including calcareous mudstone, sandstone, shale and siltstone (Text-fig. 9A–B). This break occurs at the disconformity observed in the Grassy Island section (refer to Text-fig. 2). Otherwise, it appears that *Buchia* morphology does not correlate with lithology and that substrate-related ecophenotypic variation may not be an important influence on morphology on a large scale. However, this relationship needs to be assessed further to include a more comprehensive facies analysis and also to



TEXT-FIG. 9. A, PCA results using lithology as the grouping variable from the section at Grassy Island. Axes PC1 and PC2 represent 70 per cent and 15.4 per cent, respectively, of the variation within the section. While there is a clear separation in the morphospace at the disconformity between the mudstones of the lower part of the section (Kapoose Formation) and the coarser-grained lithofacies in the upper part (One Tree Formation), shell morphology does not appear to be highly related to lithofacies, taken as a proxy for palaeoenvironment. B, PCA results, with only the group means plotted, using lithology as the grouping variable from the section at Grassy Island. Axes PC1 and PC2 represent 70 per cent and 15.4 per cent, respectively, of the variation within the section. There is no obvious trend in mean shape from coarse to fine facies.

examine other sections from different regions of British Columbia. For example, if evolutionary patterns are similar between different terranes in British Columbia, this provides better evidence that ecophenotypy is not a significant factor on morphological change. We also believe that ecophenotypic change is more important to consider when assessing patterns on smaller time scales. This research encompasses approximately 10 myr and surely any morphological change over this amount of time is owing to evolutionary (genotypic) rather than phenotypic changes.

CONCLUSIONS

Results from the multivariate techniques employed in our study have important implications for the taxonomy of the genus *Buchia* and its usefulness for biostratigraphy. This research is also particularly useful for investigation into evolutionary patterns over time because: (1) it includes abundant material from closely spaced stratigraphic intervals that provides fine-scale temporal resolution; (2) morphological change through time within the genus is captured using a standard suite of biometrics; and (3) ecophenotypic variation in shell morphology is considered.

In conclusion, we have found that: (1) phenetic discrimination using both outline shapes and angular and linear measurements appears equally useful for buchiid species discrimination; (2) existing *Buchia* species concepts in the western Canadian Cordillera are supported through the use of multivariate morphometrics, despite significant overlap in *Buchia* morphospace: this overlap

means that moderate to large sample sizes are required for biostratigraphic studies, especially when other taxonomically important features (including those of the hinge, right valve and ornament) are not preserved; (3) species variants as described by Crickmay (1925, 1930) and Jeletzky (1965) may not be taxonomically valid; (4) overall, the angles of the crest-line and inflation are the most important discriminatory variables for differentiating species of *Buchia*, although discrimination works best when comparing species pairs rather than groups and primary discriminatory variables change depending on the species pair; (5) morphological change exhibited by Grassy Island *Buchia* species from the Late Jurassic to the Early Cretaceous is gradualistic; and (6) preliminary analysis suggests that morphological change within *Buchia* does not appear to be a function of ecophenotypic variation, but this hypothesis requires further assessment.

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APPENDIX

Variable means used to characterize shell morphology for each species of Buchia

	Morphological variables									
Species	Angle J	Angle I	Ld	Lv	Wa	Wp	Dd	Dv	In	Delta
B. pacifica	−5.19	11.60	2.16	1.38	1.20	1.07	1.42	2.13	1.27	0.22
B. tolmatschowi	−13.10	13.10	2.51	1.64	1.35	1.14	1.85	2.45	1.46	0.24
B. uncitoides	−2.53	3.17	1.78	1.09	0.99	0.80	1.07	1.88	0.81	0.16
B. okensis	0.65	−0.52	1.91	0.93	1.16	0.96	1.16	2.11	0.97	0.19
B. columbiana	14.80	−12.90	1.59	1.18	0.93	0.71	0.92	1.71	0.81	0.13